NanoGT Match Report: Hemophilia A

Disease: Hemophilia A (ORPHA:98878)
Primary gene: F8
Gene CDS: 4374 bp
Inheritance: X-linked recessive
Target tissues scored: liver

Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility; AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone; Vector does not naturally cover all annotated disease tissues: liver.

Disease Mechanism Evidence

Molecular mechanism: loss of function
Mechanistic detail: Factor VIII deficiency from pathogenic F8 variants; liver-directed F8 expression is an approved gene-addition precedent (Roctavian)
Gene-addition compatibility: compatible
Preferred modality class: gene addition or factor expression
Evidence level/status: direct / needs_user_fact_check
Evidence summary: BioMarin Roctavian (valoctocogene roxaparvovec) approved for Hemophilia A; AAV5 liver-directed F8 expression precedent
Evidence source: OMIM Hemophilia A 306700

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
- Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
- Endpoint readiness: liver/metabolic targets may have biochemical biomarkers, but biomarker correction must be linked to clinical benefit.

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Hemgenix	AAV5	8.5/10	High	approved
2	Roctavian	AAV5	8.5/10	High	approved
3	SPK-8011	AAVrh10	8.1/10	High	phase3
4	DTX201	AAV8	7.8/10	High	phase2
5	Libmeldy	LV	7.3/10	Medium	approved

Match #1: Hemgenix

Precedent disease: Hemophilia B
Vector: AAV5
Tissue target: liver
Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.48**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 4374bp / cargo 4700bp (93% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Disease mechanism: loss of function — Factor VIII deficiency from pathogenic F8 variants; liver-directed F8 expression is an approved gene-addition precedent (Roctavian)
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BioMarin Roctavian (valoctocogene roxaparvovec) approved for Hemophilia A; AAV5 liver-directed F8 expression precedent

- Mechanism source: OMIM Hemophilia A 306700 (https://omim.org/entry/306700)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — F8 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #2: Roctavian

Precedent disease: Hemophilia A
Vector: AAV5
Tissue target: liver
Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.48**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 4374bp / cargo 4700bp (93% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Disease mechanism: loss of function — Factor VIII deficiency from pathogenic F8 variants; liver-directed F8 expression is an approved gene-addition precedent (Roctavian)
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BioMarin Roctavian (valoctocogene roxaparvovec) approved for Hemophilia A; AAV5 liver-directed F8 expression precedent
- Mechanism source: OMIM Hemophilia A 306700 (https://omim.org/entry/306700)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — F8 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #3: SPK-8011

Precedent disease: Hemophilia A
Vector: AAVrh10
Tissue target: liver
Composite score: 8.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**8.14**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 4374bp / cargo 4700bp (93% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Disease mechanism: loss of function — Factor VIII deficiency from pathogenic F8 variants; liver-directed F8 expression is an approved gene-addition precedent (Roctavian)
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BioMarin Roctavian (valoctocogene roxaparvovec) approved for Hemophilia A; AAV5 liver-directed F8 expression precedent

- Mechanism source: OMIM Hemophilia A 306700 (https://omim.org/entry/306700)
- Approval status: phase3
- Vector immunogenicity (AAVrh10): moderate (~10%) — significant proportion may require pre-screening or exclusion
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — F8 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: DTX201

Precedent disease: Hemophilia A
Vector: AAV8
Tissue target: liver
Composite score: 7.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.81**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 4374bp / cargo 4700bp (93% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Disease mechanism: loss of function — Factor VIII deficiency from pathogenic F8 variants; liver-directed F8 expression is an approved gene-addition precedent (Roctavian)
- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BioMarin Roctavian (valoctocogene roxaparvovec) approved for Hemophilia A; AAV5 liver-directed F8 expression precedent
- Mechanism source: OMIM Hemophilia A 306700 (https://omim.org/entry/306700)
- Approval status: phase2
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — F8 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility

- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Libmeldy

Precedent disease: Metachromatic leukodystrophy
Vector: LV
Tissue target: hematopoietic/CNS
Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Modality compatibility	2.00	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	**7.29**	**10.0**	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 4374bp / cargo 8000bp (55% utilized)
- No tissue overlap (disease: [‘liver’], vector: [‘hematopoietic’])
- Both secreted proteins — systemic delivery viable
- LOF inheritance — compatible for gene replacement
- Different pathway (coagulation vs leukodystrophy)
- Disease mechanism: loss of function — Factor VIII deficiency from pathogenic F8 variants; liver-directed F8 expression is an approved gene-addition precedent (Roctavian)

- Gene-addition modality compatibility: supports gene addition
- Mechanism evidence: BioMarin Roctavian (valoctocogene roxaparvovec) approved for Hemophilia A; AAV5 liver-directed F8 expression precedent
- Mechanism source: OMIM Hemophilia A 306700 (https://omim.org/entry/306700)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — F8 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: liver
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility